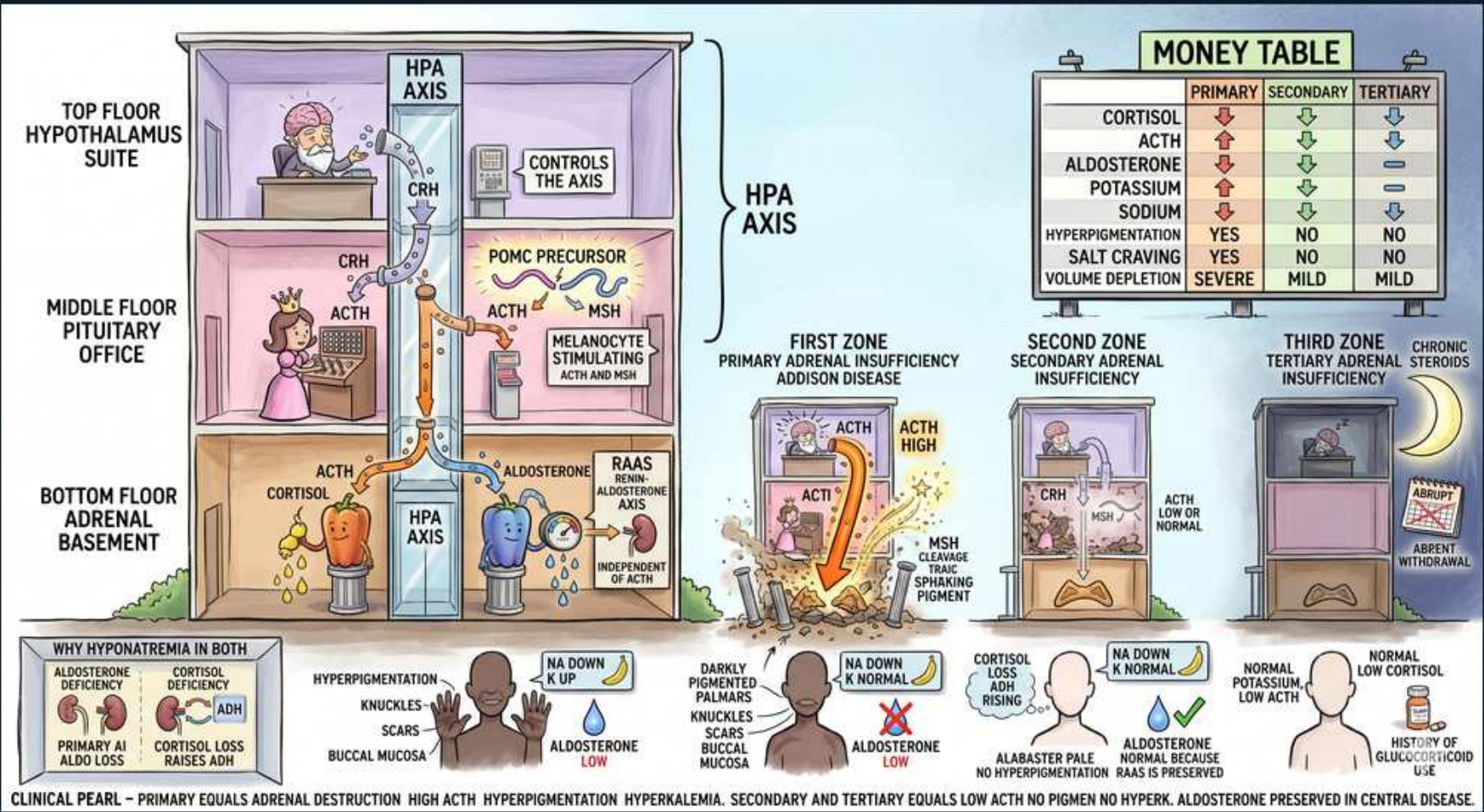


Adrenal Insufficiency — Primary vs Secondary vs Tertiary



1. HPA axis = three floors

WHAT YOU SEE

Building: hypothalamus suite / pituitary office / adrenal basement

WHAT IT ENCODES

The HPA axis runs hypothalamus (CRH) → pituitary (ACTH) → adrenal (cortisol). The level of the lesion (which floor) defines primary, secondary, or tertiary insufficiency and the lab pattern.

BOARD TRAP

Localize by ACTH and aldosterone: only the adrenal (primary) level loses aldosterone and raises ACTH.

2. CRH from hypothalamus

WHAT YOU SEE

Top floor releases CRH

WHAT IT ENCODES

CRH from the hypothalamus drives pituitary ACTH. Tertiary insufficiency = loss of CRH, most often from chronic exogenous steroids suppressing the hypothalamus.

BOARD TRAP

Tertiary AI is usually iatrogenic (chronic steroid use), not a hypothalamic tumor.

3. ACTH from pituitary

WHAT YOU SEE

Middle floor POMC precursor → ACTH and MSH

WHAT IT ENCODES

ACTH derives from POMC, which also yields melanocyte-stimulating hormone (MSH). In primary AI, high ACTH drives high MSH → hyperpigmentation. Low ACTH (secondary/tertiary) gives no pigment.

BOARD TRAP

Hyperpigmentation occurs ONLY in primary AI (high ACTH/MSH); pale skin points to secondary/tertiary.

4. Cortisol + aldosterone from adrenal

WHAT YOU SEE

Bottom floor adrenal makes cortisol and aldosterone (RAAS)

WHAT IT ENCODES

The adrenal cortex makes cortisol (ACTH-driven) and aldosterone (RAAS/renin-driven, independent of ACTH). This independence is why aldosterone is lost only in primary AI.

BOARD TRAP

Aldosterone is controlled by RAAS, not ACTH — so secondary/tertiary AI preserves aldosterone.

5. Money table — primary vs sec vs tert

WHAT YOU SEE

Top-right comparison table of cortisol, ACTH, aldosterone, Na, K

WHAT IT ENCODES

Primary: cortisol low, ACTH HIGH, aldosterone low, Na low, K high, hyperpigment +. Secondary/tertiary: cortisol low, ACTH low/normal, aldosterone normal, Na low (from ADH), K normal, no pigment.

BOARD TRAP

Hyperkalemia + hyperpigmentation + high ACTH = primary (Addison); normal K + pale = central.

6. First zone: primary (Addison)

WHAT YOU SEE

First zone panel: primary adrenal insufficiency, ACTH high

WHAT IT ENCODES

Primary AI (Addison) = destruction of the adrenal cortex (autoimmune in the West, TB worldwide). High ACTH, low cortisol AND low aldosterone → salt craving, hypotension, hyperkalemia, hyperpigmentation.

BOARD TRAP

Only primary AI loses aldosterone → hyperkalemia and salt craving; central forms do not.

7. Second zone: secondary

WHAT YOU SEE

Second zone: secondary adrenal insufficiency, ACTH low, aldosterone preserved

WHAT IT ENCODES

Secondary AI = pituitary failure of ACTH (tumor, surgery, Sheehan, apoplexy). Cortisol low, ACTH low, aldosterone NORMAL (RAAS intact). No hyperpigmentation; K normal; Na may be low from cortisol-driven ADH.

BOARD TRAP

Secondary AI does NOT cause hyperkalemia — aldosterone is preserved. Skin is pale, not pigmented.

8. Third zone: tertiary (chronic steroids)

WHAT YOU SEE

Third zone: tertiary insufficiency, history of glucocorticoid use, abrupt withdrawal

WHAT IT ENCODES

Tertiary AI = suppressed hypothalamic CRH from prolonged exogenous glucocorticoids; abrupt steroid withdrawal precipitates crisis. Labs mirror secondary (low ACTH, preserved aldosterone).

BOARD TRAP

The single most common cause of AI overall is exogenous steroid withdrawal — taper, never stop abruptly.

9. Hyponatremia in BOTH

WHAT YOU SEE

Bottom panel: why hyponatremia in both primary and secondary

WHAT IT ENCODES

Hyponatremia occurs in all forms: in primary from aldosterone loss (Na wasting) AND cortisol loss raising ADH; in secondary/tertiary purely from cortisol-deficiency-driven ADH excess. Potassium separates them.

BOARD TRAP

Low Na is common to all AI; use POTASSIUM (high in primary, normal centrally) to localize, not sodium.

10. Hyperpigmentation sites

WHAT YOU SEE

Hand/knuckle/scars/buccal mucosa darkening

WHAT IT ENCODES

Primary AI hyperpigmentation is most visible at palmar creases, knuckles, scars, and buccal mucosa due to MSH from elevated POMC/ACTH. Its presence essentially rules out central AI.

BOARD TRAP

Buccal/palmar-crease pigmentation = primary AI; its absence with low cortisol suggests central cause.

11. ADH up in primary

WHAT YOU SEE

Na down, K up arrow, ADH up in primary

WHAT IT ENCODES

In primary AI, cortisol deficiency removes tonic inhibition of ADH, so ADH rises and worsens hyponatremia on top of aldosterone-driven sodium loss.

BOARD TRAP

Cortisol loss itself raises ADH — explains hyponatremia even when aldosterone is intact (central AI).

12. Aldosterone preserved centrally

WHAT YOU SEE

Secondary panel: Na normal/low, K normal, aldosterone preserved

WHAT IT ENCODES

Because aldosterone answers to RAAS not ACTH, central AI keeps potassium and acid-base normal; only cortisol-dependent water handling is off → mild hyponatremia, normal K.

BOARD TRAP

Don't expect hyperkalemia or metabolic acidosis in secondary/tertiary AI — aldosterone protects them.

13. Tertiary cause = steroid history

WHAT YOU SEE

Moon icon: history of glucocorticoid use

WHAT IT ENCODES

A documented course of chronic glucocorticoids (oral, inhaled high-dose, intra-articular) with recent taper/stop is the cue for tertiary AI; cosyntropin testing may be falsely reassuring early.

BOARD TRAP

Recent steroid taper presenting with crisis = tertiary AI — give stress-dose hydrocortisone immediately.

14. Clinical pearl banner

WHAT YOU SEE

Bottom banner: primary = adrenal destruction, high ACTH, pigment, hyperkalemia

WHAT IT ENCODES

Pearl: primary = adrenal destruction (high ACTH, hyperpigmentation, hyperkalemia, low aldosterone). Secondary/tertiary = low ACTH, no pigment, no hyperkalemia, aldosterone preserved.

BOARD TRAP

The whole scene reduces to: ACTH direction + potassium + skin pigment localize the lesion level.